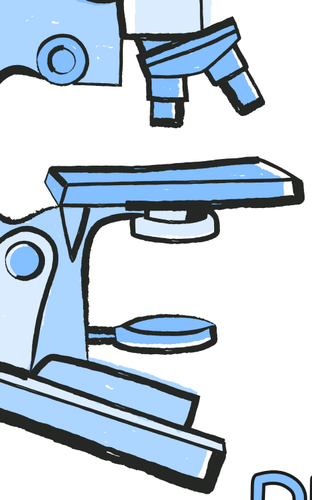
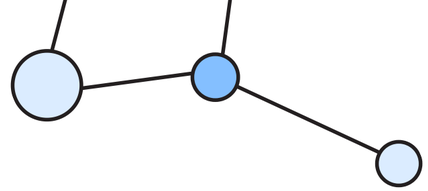
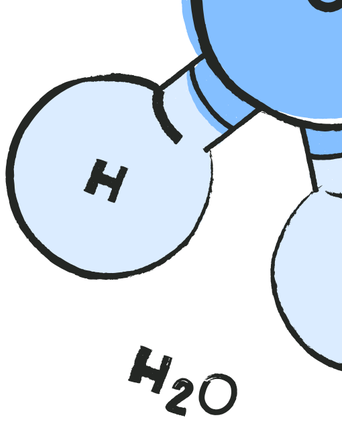


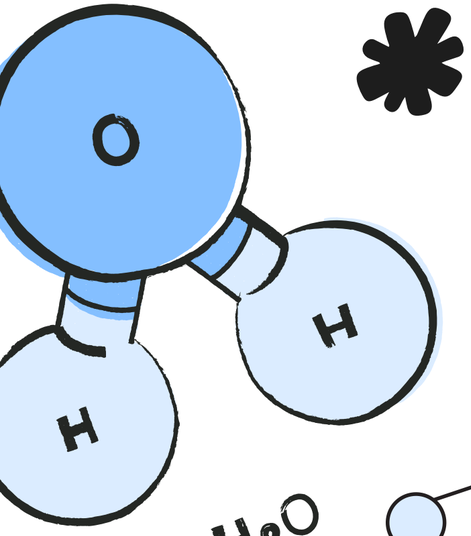


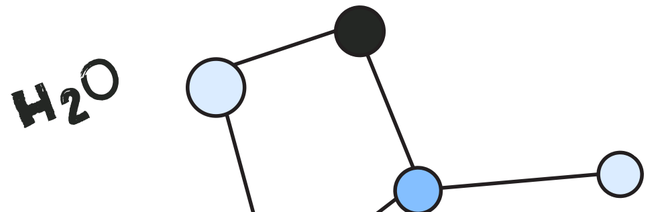


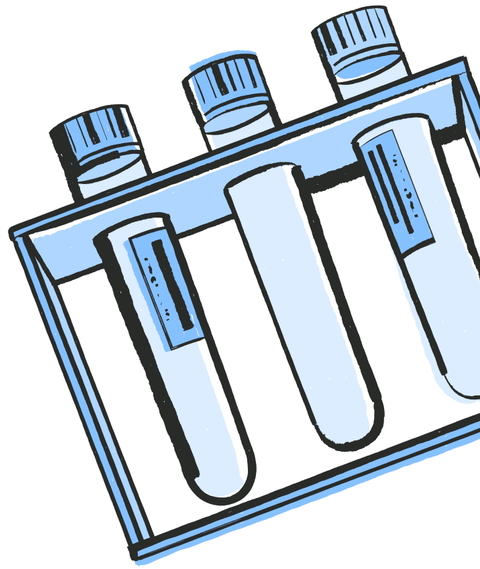
PREDICTING EXTUBATION FAILURE FROM PRE-EXTUBATION CLINICAL TRENDS

Group 25

Jiajian He 1357602
Wenhao Duan 1695089
Haoran Zhang 1697380
Xing Kun 1773853
Zongcheng Jiang 1653403







EXTUBATION REMAINS A HIGH-RISK CLINICAL DECISION

15.4%
required reintubation within 48 hours

32.6% vs 6.6% (Dadam et al., 2024)
ICU mortality among reintubated vs succesfully extubated patients

Extubation timing is critical

Extubate too early

Respiratory instability
↓
Emergency reintubation
↓
Longer ventilation and ICU stay

Extubate too late

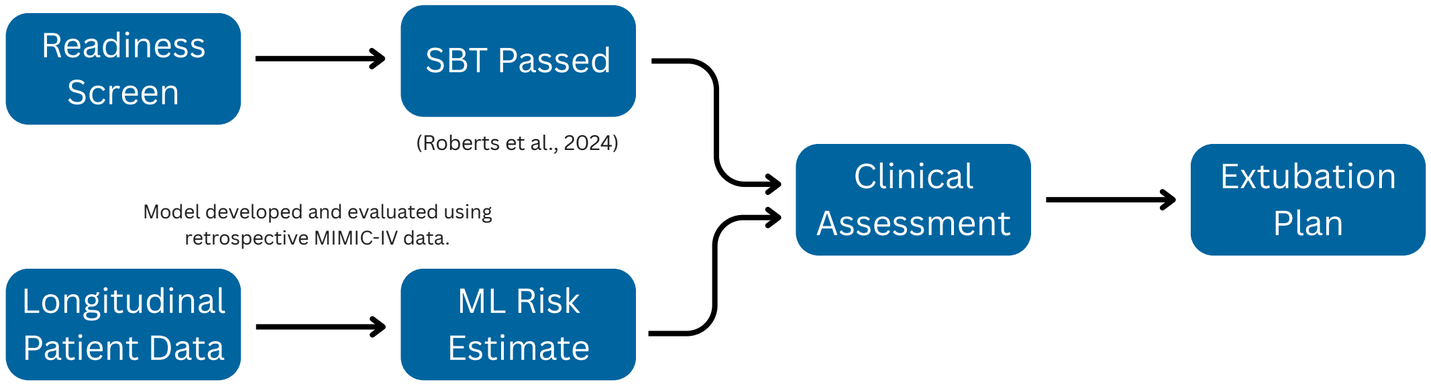
Unnecessarily prolonged mechanical ventilation
↓
Ventilator-associated pneumonia and other complications
↓
Delayed recovery

(Zhao et al., 2021)

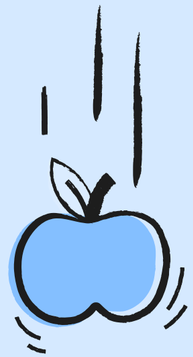
Can machine learning using pre-extubation clinical data improve prediction of 48-hour extubation failure beyond single-point clinical measurements?

Study population: *mechanically ventilated adult ICU patients.*
Extubation failure: *reintubation or death within 48 hours.*

Proposed pathway



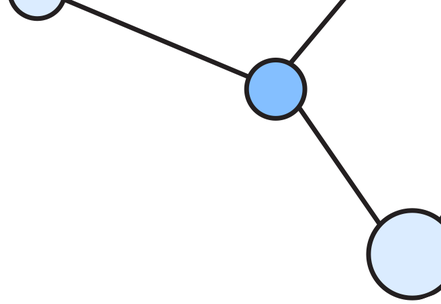
PREVIOUS WORK



Chen et al., 2019
- LightGBM for extubation failure prediction
- Structured ICU variables
- SHAP for feature interpretation

Zhao et al., 2021
- MIMIC-IV extubation failure prediction
- CatBoost using clinical/laboratory variables
- SHAP-based feature importance
- EF defined within 48 h after extubation

Bakos et al., 2026
- Systematic review of extubation ML models
- Classical ML demonstrated strong discrimination
- Limited external validation
- High heterogeneity across existing studies



What remains unclear?

Can short-term pre-extubation patterns provide additional information beyond conventional single-point measures?

Can multivariable ML improve prediction while remaining clinically interpretable?

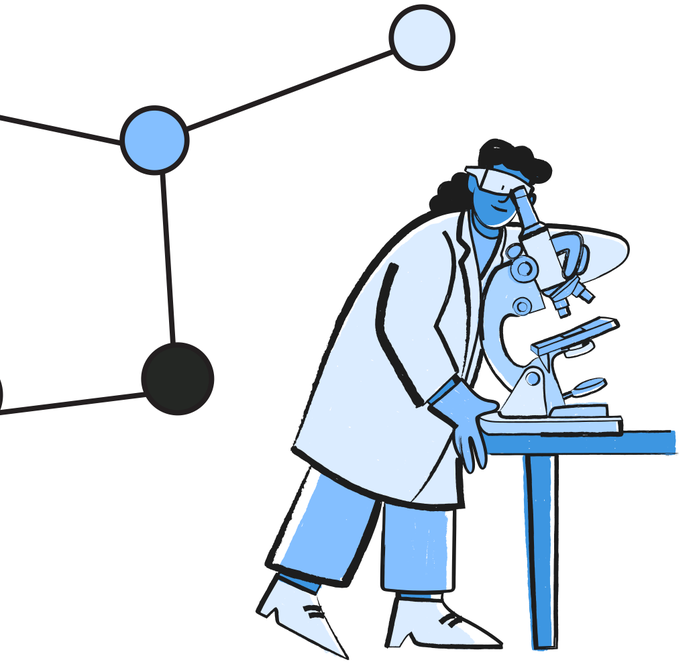
How robust are existing prediction models when applied to different patient cohorts or time periods?

DATA SOURCE



MIMIC-IV v3.1 database[Johnson et al., 2024]

- **procedureevents:**
  - extubation and reintubation timestamps for label generation
- **chartevents:**
  - respiratory parameters (RR, SpO_2, FiO_2, PEEP, tidal volume, minute ventilation)
- **labevents:**
  - arterial blood-gas results (pH, $PaCO_2$, PaO_2, HCO_3^-, lactate)
- **admissions and icustays:**
  - demographic and contextual data (age, sex, ICU type, length of stay)



DATA PROCESSING & DIGITAL PHENOTYPING

- Label:
  - reintubation ≤ 48 h after extubation or death → extubation failure
- Feature extraction:
  - 6-hour pre-extubation window
  - compute hourly averages and variability for respiratory parameters, vital signs, and blood gases
- Derived features:
  a. RSBI (RR/VT)
  b. PaO_2/FiO_2,
  c. Hemodynamic stability index
- Data cleaning:
  - merge chartevents, labevents, inputevents, icustays
  - remove duplicates and implausible timestamps
  - use forward-fill + median imputation for sparse vital signs
  - patients with >50% missing data in the 6-hour window are excluded
- Cohort selection:
  - adult ICU patients with mechanical ventilation
  - first invasive ventilation episode and its first extubation
  - ≥6 hours of valid physiological data before extubation
  - Exclusion Criteria
    - Do-Not-Intubate (DNI) or Do Not Resuscitate (DNR)
    - Unplanned extubation - patient- and non patient initiated (procedure events)
  - Expected Cohort Size:
    - around 3000 patients [Zhang et al., 2023]

ML APPROACH

Predicting extubation failure within 48 hours.
Using features from 6 h immediately before extubation, one row per patient, class weights during training.

Role	Model	Why
Baseline	Random guessing	Reference point
Existing practice	RSBI = breathing rate / breath size	Commonly used bedside index. However, it only catches 58% of failures [Trivedi 2022]
Interpretable	Logistic regression	It maintained accuracy on different hospitals' data, while other complex models did not [Zhao 2021]
Blackbox	XGBoost	Tree-based models outperform deep learning and transformers at ~10k sample size [Grinsztajn 2022]

Validation

- Nested cross-validation

Inner loop: tune hyperparameters
Outer loop: evaluate and compare performance

- Later time period held back

Train on earlier admissions and test on the most recent (using anchor_year_group).

- Missing values

Fill and scale values calculated from the training fold only

METRICS OF SUCCESS & OUTCOMES

- Primary success criterion: AUROC ≈ 0.80, based on previous extubation-failure studies.
- AUPRC should be clearly higher than the prevalence-based baseline.

Supplementary metrics: Sensitivity, Specificity

- Assess classification performance at selected thresholds and the reliability of predicted probabilities.

Interpretability: SHAP

- Identify the clinical features that contribute most strongly to model predictions.

Limitation:

- All data came from the same hospital

Metric	Expected Level	Why chosen
AUROC	High (≈0.80)	Measures overall discrimination
AUPRC	Above baseline	Handles class imbalance
Sensitivity	High	Minimise missed failures
Specificity	Moderate–High	Avoid unnecessary delay
Calibration	Good	Ensures probabilities are reliable

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THANK YOU!



AI Declaration

I acknowledge the use of Claude [claude.ai] to help me brainstorm ideas for the research topic.

It is not used for text or code generation and content creation